

GENERIC REFINERY - GULF COAST 100 kBPSP Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU		SHELL & TUBE EXCHANGER DATASHEET TEMA R / API 660 E-101 - Crude / Top pumparound		DOC. No. CFU-000-ME-DS-004 REV / DATE A / 2026-10-02 STATUS Issued for review (FEED) PREPARED CLAUDE CODE CHK: - APP: - SHEET 1	
PERFORMANCE OF ONE UNIT					
1	Size / type (TEMA)	1200-6096 AES (ID mm - tube length mm)	2	Connected in	1 parallel x 1 series
3	Surface per unit (total) / per shell	359 / 359 m2	4	Shells per unit	1
5	Service	Crude / Top pumparound	6	Heat exchanged	8.09 MW
7	MTD (corrected)	68.4 C	8	Transfer rate, service (clean: vendor)	330 W/m2K
SHELL SIDE / TUBE SIDE					
9	Fluid name (shell / tube)	Crude / Top pumparound	10	Total flow (shell / tube)	569.9 / 166.8 t/h
11	Temperature in (shell / tube)	30 / 153 C	12	Temperature out (shell / tube)	56 / 83 C
13	Phase	L / L	14	Physical properties	Per H&MB / vendor simulation
15	Allowable pressure drop (shell / tube)	0.7 / 0.7 bar	16	Fouling resistance (shell / tube)	0.0005 / 0.0004 m2K/W
CONSTRUCTION OF ONE SHELL					
17	Design pressure (shell / tube)	30 / 20 barg	18	Design temperature	185 C
19	Test pressure (shell / tube)	39.0 / 26.0 barg	20	Passes per shell (shell / tube)	1 / 4
21	Corrosion allowance	3.0 mm	22	Connections in/out (shell)	16" 300#
23	Connections in/out (tube)	8" 300#	24	Tube no. / OD / thk / length	756 / 25.4 mm / 12 BWG (2.77 mm) / 6096 mm
25	Tube pitch / layout	31.75 mm / 90 deg (square, cleanable)	26	Tube material	CS tubes
27	Shell material / ID / thk	SA-516-70 / 1200 mm / 20 mm	28	Channel / bonnet thickness	14 mm
29	Tubesheet	Vendor (TEMA R / ASME UHX)	30	Floating head	Split-ring (S)
31	Baffles	Single segmental, cut 25 % (V) - spacing vendor	32	Impingement protection	Plate (rho.v2 > 2230)
33	Expansion joint	None	34	Gaskets	Double-jacketed / spiral-wound
35	Code requirements	ASME VIII-1 / TEMA R / API 660	36	Bundle pull / removable	Yes
37	Weight per shell empty / water filled	18.9 / 27.2 t	38	Bundle weight (est.)	9.6 t
NOTES					
1. TEMA datasheet layout. Thermal design (HTRI) by vendor - geometry here is an estimate from area and the tube-count correlation. 2. Crude shell-side exchangers: square pitch, removable bundle, 1 m min. tube velocity (tube side) / no dead zones. 3. Stacking: max. 2 shells high; bundle-pull clearance = tube length + 1.5 m.					

GENERIC REFINERY - GULF COAST 100 kBPSD Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU		SHELL & TUBE EXCHANGER DATASHEET TEMA R / API 660 E-102 - Crude / Kerosene product		DOC. No. CFU-000-ME-DS-004	
				REV / DATE A / 2026-10-02	
				STATUS Issued for review (FEED)	
				PREPARED CLAUDE CODE CHK: - APP: -	
				SHEET 2	
PERFORMANCE OF ONE UNIT					
1	Size / type (TEMA)	1300-6096 AES (ID mm - tube length mm)	2	Connected in	1 parallel x 1 series
3	Surface per unit (total) / per shell	438 / 438 m2	4	Shells per unit	1
5	Service	Crude / Kerosene product	6	Heat exchanged	6.55 MW
7	MTD (corrected)	49.9 C	8	Transfer rate, service (clean: vendor)	300 W/m2K
SHELL SIDE / TUBE SIDE					
9	Fluid name (shell / tube)	Crude / Kerosene product	10	Total flow (shell / tube)	569.9 / 68.6 t/h
11	Temperature in (shell / tube)	56 / 212 C	12	Temperature out (shell / tube)	75 / 76 C
13	Phase	L / L	14	Physical properties	Per H&MB / vendor simulation
15	Allowable pressure drop (shell / tube)	0.7 / 0.7 bar	16	Fouling resistance (shell / tube)	0.0005 / 0.0004 m2K/W
CONSTRUCTION OF ONE SHELL					
17	Design pressure (shell / tube)	30 / 20 barg	18	Design temperature	240 C
19	Test pressure (shell / tube)	39.0 / 26.0 barg	20	Passes per shell (shell / tube)	1 / 4
21	Corrosion allowance	3.0 mm	22	Connections in/out (shell)	16" 300#
23	Connections in/out (tube)	6" 300#	24	Tube no. / OD / thk / length	923 / 25.4 mm / 12 BWG (2.77 mm) / 6096 mm
25	Tube pitch / layout	31.75 mm / 90 deg (square, cleanable)	26	Tube material	CS tubes
27	Shell material / ID / thk	SA-516-70 / 1300 mm / 20 mm	28	Channel / bonnet thickness	16 mm
29	Tubesheet	Vendor (TEMA R / ASME UHX)	30	Floating head	Split-ring (S)
31	Baffles	Single segmental, cut 25 % (V) - spacing vendor	32	Impingement protection	Plate (rho.v2 > 2230)
33	Expansion joint	None	34	Gaskets	Double-jacketed / spiral-wound
35	Code requirements	ASME VIII-1 / TEMA R / API 660	36	Bundle pull / removable	Yes
37	Weight per shell empty / water filled	22.4 / 32.0 t	38	Bundle weight (est.)	11.8 t
NOTES					
1. TEMA datasheet layout. Thermal design (HTRI) by vendor - geometry here is an estimate from area and the tube-count correlation.					
2. Crude shell-side exchangers: square pitch, removable bundle, 1 m min. tube velocity (tube side) / no dead zones.					
3. Stacking: max. 2 shells high; bundle-pull clearance = tube length + 1.5 m.					

GENERIC REFINERY - GULF COAST 100 kBPST Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU		SHELL & TUBE EXCHANGER DATASHEET TEMA R / API 660 E-103 - Crude / LVGO pumparound/product		DOC. No. CFU-000-ME-DS-004 REV / DATE A / 2026-10-02 STATUS Issued for review (FEED) PREPARED CLAUDE CODE CHK: - APP: - SHEET 3	
PERFORMANCE OF ONE UNIT					
1	Size / type (TEMA)	1250-6096 AES (ID mm - tube length mm)	2	Connected in	1 parallel x 2 series
3	Surface per unit (total) / per shell	804 / 402 m2	4	Shells per unit	2
5	Service	Crude / LVGO pumparound/product	6	Heat exchanged	11.45 MW
7	MTD (corrected)	50.9 C	8	Transfer rate, service (clean: vendor)	280 W/m2K
SHELL SIDE / TUBE SIDE					
9	Fluid name (shell / tube)	Crude / LVGO pumparound/product	10	Total flow (shell / tube)	569.9 / 132.9 t/h
11	Temperature in (shell / tube)	75 / 224 C	12	Temperature out (shell / tube)	108 / 95 C
13	Phase	L / L	14	Physical properties	Per H&MB / vendor simulation
15	Allowable pressure drop (shell / tube)	0.7 / 0.7 bar	16	Fouling resistance (shell / tube)	0.0005 / 0.0004 m2K/W
CONSTRUCTION OF ONE SHELL					
17	Design pressure (shell / tube)	30 / 20 barg	18	Design temperature	255 C
19	Test pressure (shell / tube)	39.0 / 26.0 barg	20	Passes per shell (shell / tube)	1 / 4
21	Corrosion allowance	3.0 mm	22	Connections in/out (shell)	16" 300#
23	Connections in/out (tube)	8" 300#	24	Tube no. / OD / thk / length	848 / 25.4 mm / 12 BWG (2.77 mm) / 6096 mm
25	Tube pitch / layout	31.75 mm / 90 deg (square, cleanable)	26	Tube material	CS tubes
27	Shell material / ID / thk	SA-516-70 / 1250 mm / 20 mm	28	Channel / bonnet thickness	14 mm
29	Tubesheet	Vendor (TEMA R / ASME UHX)	30	Floating head	Split-ring (S)
31	Baffles	Single segmental, cut 25 % (V) - spacing vendor	32	Impingement protection	Plate (rho.v2 > 2230)
33	Expansion joint	None	34	Gaskets	Double-jacketed / spiral-wound
35	Code requirements	ASME VIII-1 / TEMA R / API 660	36	Bundle pull / removable	Yes
37	Weight per shell empty / water filled	20.7 / 29.6 t	38	Bundle weight (est.)	10.8 t
NOTES					
1. TEMA datasheet layout. Thermal design (HTRI) by vendor - geometry here is an estimate from area and the tube-count correlation. 2. Crude shell-side exchangers: square pitch, removable bundle, 1 m min. tube velocity (tube side) / no dead zones. 3. Stacking: max. 2 shells high; bundle-pull clearance = tube length + 1.5 m.					

GENERIC REFINERY - GULF COAST 100 kBPSD Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU		SHELL & TUBE EXCHANGER DATASHEET TEMA R / API 660 E-104 - Crude / Diesel product		DOC. No. CFU-000-ME-DS-004	
				REV / DATE A / 2026-10-02	
				STATUS Issued for review (FEED)	
				PREPARED CLAUDE CODE CHK: - APP: -	
				SHEET 4	
PERFORMANCE OF ONE UNIT					
1	Size / type (TEMA)	1200-6096 AES (ID mm - tube length mm)	2	Connected in	1 parallel x 1 series
3	Surface per unit (total) / per shell	378 / 378 m2	4	Shells per unit	1
5	Service	Crude / Diesel product	6	Heat exchanged	4.44 MW
7	MTD (corrected)	40.5 C	8	Transfer rate, service (clean: vendor)	290 W/m2K
SHELL SIDE / TUBE SIDE					
9	Fluid name (shell / tube)	Crude / Diesel product	10	Total flow (shell / tube)	569.9 / 81.6 t/h
11	Temperature in (shell / tube)	108 / 205 C	12	Temperature out (shell / tube)	121 / 128 C
13	Phase	L / L	14	Physical properties	Per H&MB / vendor simulation
15	Allowable pressure drop (shell / tube)	0.7 / 0.7 bar	16	Fouling resistance (shell / tube)	0.0005 / 0.0004 m2K/W
CONSTRUCTION OF ONE SHELL					
17	Design pressure (shell / tube)	30 / 20 barg	18	Design temperature	235 C
19	Test pressure (shell / tube)	39.0 / 26.0 barg	20	Passes per shell (shell / tube)	1 / 4
21	Corrosion allowance	3.0 mm	22	Connections in/out (shell)	16" 300#
23	Connections in/out (tube)	6" 300#	24	Tube no. / OD / thk / length	797 / 25.4 mm / 12 BWG (2.77 mm) / 6096 mm
25	Tube pitch / layout	31.75 mm / 90 deg (square, cleanable)	26	Tube material	CS tubes
27	Shell material / ID / thk	SA-516-70 / 1200 mm / 20 mm	28	Channel / bonnet thickness	14 mm
29	Tubesheet	Vendor (TEMA R / ASME UHX)	30	Floating head	Split-ring (S)
31	Baffles	Single segmental, cut 25 % (V) - spacing vendor	32	Impingement protection	Plate (rho.v2 > 2230)
33	Expansion joint	None	34	Gaskets	Double-jacketed / spiral-wound
35	Code requirements	ASME VIII-1 / TEMA R / API 660	36	Bundle pull / removable	Yes
37	Weight per shell empty / water filled	19.5 / 27.7 t	38	Bundle weight (est.)	10.2 t
NOTES					
1. TEMA datasheet layout. Thermal design (HTRI) by vendor - geometry here is an estimate from area and the tube-count correlation.					
2. Crude shell-side exchangers: square pitch, removable bundle, 1 m min. tube velocity (tube side) / no dead zones.					
3. Stacking: max. 2 shells high; bundle-pull clearance = tube length + 1.5 m.					

GENERIC REFINERY - GULF COAST 100 kBPST Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU		SHELL & TUBE EXCHANGER DATASHEET TEMA R / API 660 E-105 - Crude / Vacuum residue		DOC. No. CFU-000-ME-DS-004	
				REV / DATE A / 2026-10-02	
				STATUS Issued for review (FEED)	
				PREPARED CLAUDE CODE CHK: - APP: -	
				SHEET 5	
PERFORMANCE OF ONE UNIT					
1	Size / type (TEMA)	1300-6096 AES (ID mm - tube length mm)	2	Connected in	1 parallel x 1 series
3	Surface per unit (total) / per shell	431 / 431 m2	4	Shells per unit	1
5	Service	Crude / Vacuum residue	6	Heat exchanged	5.84 MW
7	MTD (corrected)	79.7 C	8	Transfer rate, service (clean: vendor)	170 W/m2K
SHELL SIDE / TUBE SIDE					
9	Fluid name (shell / tube)	Vacuum residue / Crude	10	Total flow (shell / tube)	108.3 / 569.9 t/h
11	Temperature in (shell / tube)	255 / 121 C	12	Temperature out (shell / tube)	175 / 136 C
13	Phase	L / L	14	Physical properties	Per H&MB / vendor simulation
15	Allowable pressure drop (shell / tube)	0.7 / 0.7 bar	16	Fouling resistance (shell / tube)	0.0009 / 0.0005 m2K/W
CONSTRUCTION OF ONE SHELL					
17	Design pressure (shell / tube)	30 / 20 barg	18	Design temperature	285 C
19	Test pressure (shell / tube)	39.0 / 26.0 barg	20	Passes per shell (shell / tube)	1 / 4
21	Corrosion allowance	3.0 mm	22	Connections in/out (shell)	8" 300#
23	Connections in/out (tube)	16" 300#	24	Tube no. / OD / thk / length	910 / 25.4 mm / 12 BWG (2.77 mm) / 6096 mm
25	Tube pitch / layout	31.75 mm / 90 deg (square, cleanable)	26	Tube material	2Mo tubes
27	Shell material / ID / thk	SA-516-70 / 1300 mm / 22 mm	28	Channel / bonnet thickness	16 mm
29	Tubesheet	Vendor (TEMA R / ASME UHX)	30	Floating head	Split-ring (S)
31	Baffles	Single segmental, cut 25 % (V) - spacing vendor	32	Impingement protection	Plate (rho.v2 > 2230)
33	Expansion joint	None	34	Gaskets	Double-jacketed / spiral-wound
35	Code requirements	ASME VIII-1 / TEMA R / API 660	36	Bundle pull / removable	Yes
37	Weight per shell empty / water filled	22.7 / 32.3 t	38	Bundle weight (est.)	11.6 t
NOTES					
1. TEMA datasheet layout. Thermal design (HTRI) by vendor - geometry here is an estimate from area and the tube-count correlation.					
2. Crude shell-side exchangers: square pitch, removable bundle, 1 m min. tube velocity (tube side) / no dead zones.					
3. Stacking: max. 2 shells high; bundle-pull clearance = tube length + 1.5 m.					

GENERIC REFINERY - GULF COAST 100 kBPSP Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU		SHELL & TUBE EXCHANGER DATASHEET TEMA R / API 660 E-106 - Crude / Middle (kero) pumparound		DOC. No. CFU-000-ME-DS-004 REV / DATE A / 2026-10-02 STATUS Issued for review (FEED) PREPARED CLAUDE CODE CHK: - APP: - SHEET 6	
PERFORMANCE OF ONE UNIT					
1	Size / type (TEMA)	1350-6096 AES (ID mm - tube length mm)	2	Connected in	1 parallel x 2 series
3	Surface per unit (total) / per shell	1001 / 500 m2	4	Shells per unit	2
5	Service	Crude / Middle (kero) pumparound	6	Heat exchanged	12.85 MW
7	MTD (corrected)	40.1 C	8	Transfer rate, service (clean: vendor)	320 W/m2K
SHELL SIDE / TUBE SIDE					
9	Fluid name (shell / tube)	Crude / Middle (kero) pumparound	10	Total flow (shell / tube)	569.2 / 208.3 t/h
11	Temperature in (shell / tube)	136 / 240 C	12	Temperature out (shell / tube)	170 / 160 C
13	Phase	L / L	14	Physical properties	Per H&MB / vendor simulation
15	Allowable pressure drop (shell / tube)	0.7 / 0.7 bar	16	Fouling resistance (shell / tube)	0.0005 / 0.0004 m2K/W
CONSTRUCTION OF ONE SHELL					
17	Design pressure (shell / tube)	30 / 20 barg	18	Design temperature	270 C
19	Test pressure (shell / tube)	39.0 / 26.0 barg	20	Passes per shell (shell / tube)	1 / 4
21	Corrosion allowance	3.0 mm	22	Connections in/out (shell)	16" 300#
23	Connections in/out (tube)	10" 300#	24	Tube no. / OD / thk / length	1055 / 25.4 mm / 12 BWG (2.77 mm) / 6096 mm
25	Tube pitch / layout	31.75 mm / 90 deg (square, cleanable)	26	Tube material	2Mo tubes
27	Shell material / ID / thk	SA-516-70 / 1350 mm / 22 mm	28	Channel / bonnet thickness	16 mm
29	Tubesheet	Vendor (TEMA R / ASME UHX)	30	Floating head	Split-ring (S)
31	Baffles	Single segmental, cut 25 % (V) - spacing vendor	32	Impingement protection	Plate (rho.v2 > 2230)
33	Expansion joint	None	34	Gaskets	Double-jacketed / spiral-wound
35	Code requirements	ASME VIII-1 / TEMA R / API 660	36	Bundle pull / removable	Yes
37	Weight per shell empty / water filled	25.2 / 35.6 t	38	Bundle weight (est.)	13.5 t
NOTES					
1. TEMA datasheet layout. Thermal design (HTRI) by vendor - geometry here is an estimate from area and the tube-count correlation. 2. Crude shell-side exchangers: square pitch, removable bundle, 1 m min. tube velocity (tube side) / no dead zones. 3. Stacking: max. 2 shells high; bundle-pull clearance = tube length + 1.5 m.					

GENERIC REFINERY - GULF COAST 100 kBPST Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU		SHELL & TUBE EXCHANGER DATASHEET TEMA R / API 660 E-107 - Crude / Diesel product		DOC. No. CFU-000-ME-DS-004	
				REV / DATE A / 2026-10-02	
				STATUS Issued for review (FEED)	
				PREPARED CLAUDE CODE CHK: - APP: -	
				SHEET 7	
PERFORMANCE OF ONE UNIT					
1	Size / type (TEMA)	1150-6096 AES (ID mm - tube length mm)	2	Connected in	1 parallel x 1 series
3	Surface per unit (total) / per shell	324 / 324 m2	4	Shells per unit	1
5	Service	Crude / Diesel product	6	Heat exchanged	5.83 MW
7	MTD (corrected)	62.0 C	8	Transfer rate, service (clean: vendor)	290 W/m2K
SHELL SIDE / TUBE SIDE					
9	Fluid name (shell / tube)	Crude / Diesel product	10	Total flow (shell / tube)	569.2 / 81.6 t/h
11	Temperature in (shell / tube)	170 / 294 C	12	Temperature out (shell / tube)	184 / 205 C
13	Phase	L / L	14	Physical properties	Per H&MB / vendor simulation
15	Allowable pressure drop (shell / tube)	0.7 / 0.7 bar	16	Fouling resistance (shell / tube)	0.0005 / 0.0004 m2K/W
CONSTRUCTION OF ONE SHELL					
17	Design pressure (shell / tube)	30 / 20 barg	18	Design temperature	325 C
19	Test pressure (shell / tube)	39.0 / 26.0 barg	20	Passes per shell (shell / tube)	1 / 4
21	Corrosion allowance	6.0 mm	22	Connections in/out (shell)	16" 300#
23	Connections in/out (tube)	6" 300#	24	Tube no. / OD / thk / length	683 / 25.4 mm / 12 BWG (2.77 mm) / 6096 mm
25	Tube pitch / layout	31.75 mm / 90 deg (square, cleanable)	26	Tube material	2Mo tubes
27	Shell material / ID / thk	SA-516-70 / 1150 mm / 22 mm	28	Channel / bonnet thickness	18 mm
29	Tubesheet	Vendor (TEMA R / ASME UHX)	30	Floating head	Split-ring (S)
31	Baffles	Single segmental, cut 25 % (V) - spacing vendor	32	Impingement protection	Plate (rho.v2 > 2230)
33	Expansion joint	None	34	Gaskets	Double-jacketed / spiral-wound
35	Code requirements	ASME VIII-1 / TEMA R / API 660	36	Bundle pull / removable	Yes
37	Weight per shell empty / water filled	18.2 / 25.8 t	38	Bundle weight (est.)	8.7 t
NOTES					
1. TEMA datasheet layout. Thermal design (HTRI) by vendor - geometry here is an estimate from area and the tube-count correlation.					
2. Crude shell-side exchangers: square pitch, removable bundle, 1 m min. tube velocity (tube side) / no dead zones.					
3. Stacking: max. 2 shells high; bundle-pull clearance = tube length + 1.5 m.					

GENERIC REFINERY - GULF COAST 100 kBPST Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU		SHELL & TUBE EXCHANGER DATASHEET TEMA R / API 660 E-108 - Crude / HVGO pumparound/product		DOC. No. CFU-000-ME-DS-004 REV / DATE A / 2026-10-02 STATUS Issued for review (FEED) PREPARED CLAUDE CODE CHK: - APP: - SHEET 8	
PERFORMANCE OF ONE UNIT					
1	Size / type (TEMA)	1400-6096 AES (ID mm - tube length mm)	2	Connected in	2 parallel x 2 series
3	Surface per unit (total) / per shell	2196 / 549 m2	4	Shells per unit	4
5	Service	Crude / HVGO pumparound/product	6	Heat exchanged	21.14 MW
7	MTD (corrected)	38.5 C	8	Transfer rate, service (clean: vendor)	250 W/m2K
SHELL SIDE / TUBE SIDE					
9	Fluid name (shell / tube)	Crude / HVGO pumparound/product	10	Total flow (shell / tube)	569.2 / 251.8 t/h
11	Temperature in (shell / tube)	184 / 315 C	12	Temperature out (shell / tube)	234 / 205 C
13	Phase	L / L	14	Physical properties	Per H&MB / vendor simulation
15	Allowable pressure drop (shell / tube)	0.7 / 0.7 bar	16	Fouling resistance (shell / tube)	0.0005 / 0.0004 m2K/W
CONSTRUCTION OF ONE SHELL					
17	Design pressure (shell / tube)	30 / 20 barg	18	Design temperature	345 C
19	Test pressure (shell / tube)	39.0 / 26.0 barg	20	Passes per shell (shell / tube)	1 / 4
21	Corrosion allowance	6.0 mm	22	Connections in/out (shell)	16" 300#
23	Connections in/out (tube)	10" 300#	24	Tube no. / OD / thk / length	1157 / 25.4 mm / 12 BWG (2.77 mm) / 6096 mm
25	Tube pitch / layout	31.75 mm / 90 deg (square, cleanable)	26	Tube material	9Cr-1Mo tubes
27	Shell material / ID / thk	SA-387-11-2 / 1400 mm / 25 mm	28	Channel / bonnet thickness	18 mm
29	Tubesheet	Vendor (TEMA R / ASME UHX)	30	Floating head	Split-ring (S)
31	Baffles	Single segmental, cut 25 % (V) - spacing vendor	32	Impingement protection	Plate (rho.v2 > 2230)
33	Expansion joint	None	34	Gaskets	Double-jacketed / spiral-wound
35	Code requirements	ASME VIII-1 / TEMA R / API 660	36	Bundle pull / removable	Yes
37	Weight per shell empty / water filled	28.2 / 39.3 t	38	Bundle weight (est.)	14.8 t
NOTES					
1. TEMA datasheet layout. Thermal design (HTRI) by vendor - geometry here is an estimate from area and the tube-count correlation. 2. Crude shell-side exchangers: square pitch, removable bundle, 1 m min. tube velocity (tube side) / no dead zones. 3. Stacking: max. 2 shells high; bundle-pull clearance = tube length + 1.5 m.					

GENERIC REFINERY - GULF COAST 100 kBPST Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU		SHELL & TUBE EXCHANGER DATASHEET TEMA R / API 660 E-109 - Crude / AGO product		DOC. No. CFU-000-ME-DS-004 REV / DATE A / 2026-10-02 STATUS Issued for review (FEED) PREPARED CLAUDE CODE CHK: - APP: - SHEET 9	
PERFORMANCE OF ONE UNIT					
1	Size / type (TEMA)	1000-6096 AES (ID mm - tube length mm)	2	Connected in	1 parallel x 1 series
3	Surface per unit (total) / per shell	263 / 263 m2	4	Shells per unit	1
5	Service	Crude / AGO product	6	Heat exchanged	3.11 MW
7	MTD (corrected)	45.4 C	8	Transfer rate, service (clean: vendor)	260 W/m2K
SHELL SIDE / TUBE SIDE					
9	Fluid name (shell / tube)	Crude / AGO product	10	Total flow (shell / tube)	569.2 / 45.8 t/h
11	Temperature in (shell / tube)	234 / 335 C	12	Temperature out (shell / tube)	241 / 254 C
13	Phase	L / L	14	Physical properties	Per H&MB / vendor simulation
15	Allowable pressure drop (shell / tube)	0.7 / 0.7 bar	16	Fouling resistance (shell / tube)	0.0005 / 0.0004 m2K/W
CONSTRUCTION OF ONE SHELL					
17	Design pressure (shell / tube)	30 / 20 barg	18	Design temperature	365 C
19	Test pressure (shell / tube)	39.0 / 26.0 barg	20	Passes per shell (shell / tube)	1 / 2
21	Corrosion allowance	6.0 mm	22	Connections in/out (shell)	16" 300#
23	Connections in/out (tube)	6" 300#	24	Tube no. / OD / thk / length	555 / 25.4 mm / 12 BWG (2.77 mm) / 6096 mm
25	Tube pitch / layout	31.75 mm / 90 deg (square, cleanable)	26	Tube material	9Cr-1Mo tubes
27	Shell material / ID / thk	SA-387-11-2 / 1000 mm / 20 mm	28	Channel / bonnet thickness	16 mm
29	Tubesheet	Vendor (TEMA R / ASME UHX)	30	Floating head	Split-ring (S)
31	Baffles	Single segmental, cut 25 % (V) - spacing vendor	32	Impingement protection	Plate (rho.v2 > 2230)
33	Expansion joint	None	34	Gaskets	Double-jacketed / spiral-wound
35	Code requirements	ASME VIII-1 / TEMA R / API 660	36	Bundle pull / removable	Yes
37	Weight per shell empty / water filled	14.5 / 20.2 t	38	Bundle weight (est.)	7.1 t
NOTES					
1. TEMA datasheet layout. Thermal design (HTRI) by vendor - geometry here is an estimate from area and the tube-count correlation. 2. Crude shell-side exchangers: square pitch, removable bundle, 1 m min. tube velocity (tube side) / no dead zones. 3. Stacking: max. 2 shells high; bundle-pull clearance = tube length + 1.5 m.					

GENERIC REFINERY - GULF COAST 100 kBPSD Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU		SHELL & TUBE EXCHANGER DATASHEET TEMA R / API 660 E-110 - Crude / Bottom (diesel) pumparound		DOC. No. CFU-000-ME-DS-004	
				REV / DATE A / 2026-10-02	
				STATUS Issued for review (FEED)	
				PREPARED CLAUDE CODE CHK: - APP: -	
				SHEET 10	
PERFORMANCE OF ONE UNIT					
1	Size / type (TEMA)	1300-6096 AES (ID mm - tube length mm)	2	Connected in	2 parallel x 1 series
3	Surface per unit (total) / per shell	886 / 443 m2	4	Shells per unit	2
5	Service	Crude / Bottom (diesel) pumparound	6	Heat exchanged	7.46 MW
7	MTD (corrected)	30.1 C	8	Transfer rate, service (clean: vendor)	280 W/m2K
SHELL SIDE / TUBE SIDE					
9	Fluid name (shell / tube)	Crude / Bottom (diesel) pumparound	10	Total flow (shell / tube)	569.2 / 165.7 t/h
11	Temperature in (shell / tube)	241 / 314 C	12	Temperature out (shell / tube)	258 / 261 C
13	Phase	L / L	14	Physical properties	Per H&MB / vendor simulation
15	Allowable pressure drop (shell / tube)	0.7 / 0.7 bar	16	Fouling resistance (shell / tube)	0.0005 / 0.0004 m2K/W
CONSTRUCTION OF ONE SHELL					
17	Design pressure (shell / tube)	30 / 20 barg	18	Design temperature	345 C
19	Test pressure (shell / tube)	39.0 / 26.0 barg	20	Passes per shell (shell / tube)	1 / 4
21	Corrosion allowance	6.0 mm	22	Connections in/out (shell)	16" 300#
23	Connections in/out (tube)	8" 300#	24	Tube no. / OD / thk / length	934 / 25.4 mm / 12 BWG (2.77 mm) / 6096 mm
25	Tube pitch / layout	31.75 mm / 90 deg (square, cleanable)	26	Tube material	9Cr-1Mo tubes
27	Shell material / ID / thk	SA-387-11-2 / 1300 mm / 22 mm	28	Channel / bonnet thickness	18 mm
29	Tubesheet	Vendor (TEMA R / ASME UHX)	30	Floating head	Split-ring (S)
31	Baffles	Single segmental, cut 25 % (V) - spacing vendor	32	Impingement protection	Plate (rho.v2 > 2230)
33	Expansion joint	None	34	Gaskets	Double-jacketed / spiral-wound
35	Code requirements	ASME VIII-1 / TEMA R / API 660	36	Bundle pull / removable	Yes
37	Weight per shell empty / water filled	23.2 / 32.9 t	38	Bundle weight (est.)	11.9 t
NOTES					
1. TEMA datasheet layout. Thermal design (HTRI) by vendor - geometry here is an estimate from area and the tube-count correlation.					
2. Crude shell-side exchangers: square pitch, removable bundle, 1 m min. tube velocity (tube side) / no dead zones.					
3. Stacking: max. 2 shells high; bundle-pull clearance = tube length + 1.5 m.					

GENERIC REFINERY - GULF COAST 100 kBPST Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU		SHELL & TUBE EXCHANGER DATASHEET TEMA R / API 660 E-111 - Crude / Vacuum residue		DOC. No. CFU-000-ME-DS-004	
				REV / DATE A / 2026-10-02	
				STATUS Issued for review (FEED)	
				PREPARED CLAUDE CODE CHK: - APP: -	
				SHEET 11	
PERFORMANCE OF ONE UNIT					
1	Size / type (TEMA)	1400-6096 AES (ID mm - tube length mm)	2	Connected in	2 parallel x 1 series
3	Surface per unit (total) / per shell	1094 / 547 m2	4	Shells per unit	2
5	Service	Crude / Vacuum residue	6	Heat exchanged	7.58 MW
7	MTD (corrected)	40.7 C	8	Transfer rate, service (clean: vendor)	170 W/m2K
SHELL SIDE / TUBE SIDE					
9	Fluid name (shell / tube)	Vacuum residue / Crude	10	Total flow (shell / tube)	108.3 / 569.2 t/h
11	Temperature in (shell / tube)	366 / 258 C	12	Temperature out (shell / tube)	278 / 274 C
13	Phase	L / L	14	Physical properties	Per H&MB / vendor simulation
15	Allowable pressure drop (shell / tube)	0.7 / 0.7 bar	16	Fouling resistance (shell / tube)	0.0009 / 0.0005 m2K/W
CONSTRUCTION OF ONE SHELL					
17	Design pressure (shell / tube)	30 / 20 barg	18	Design temperature	395 C
19	Test pressure (shell / tube)	39.0 / 26.0 barg	20	Passes per shell (shell / tube)	1 / 4
21	Corrosion allowance	6.0 mm	22	Connections in/out (shell)	8" 300#
23	Connections in/out (tube)	16" 300#	24	Tube no. / OD / thk / length	1154 / 25.4 mm / 12 BWG (2.77 mm) / 6096 mm
25	Tube pitch / layout	31.75 mm / 90 deg (square, cleanable)	26	Tube material	9Cr-1Mo tubes
27	Shell material / ID / thk	SA-387-11-2 / 1400 mm / 25 mm	28	Channel / bonnet thickness	18 mm
29	Tubesheet	Vendor (TEMA R / ASME UHX)	30	Floating head	Split-ring (S)
31	Baffles	Single segmental, cut 25 % (V) - spacing vendor	32	Impingement protection	Plate (rho.v2 > 2230)
33	Expansion joint	None	34	Gaskets	Double-jacketed / spiral-wound
35	Code requirements	ASME VIII-1 / TEMA R / API 660	36	Bundle pull / removable	Yes
37	Weight per shell empty / water filled	28.1 / 39.3 t	38	Bundle weight (est.)	14.7 t
NOTES					
1. TEMA datasheet layout. Thermal design (HTRI) by vendor - geometry here is an estimate from area and the tube-count correlation.					
2. Crude shell-side exchangers: square pitch, removable bundle, 1 m min. tube velocity (tube side) / no dead zones.					
3. Stacking: max. 2 shells high; bundle-pull clearance = tube length + 1.5 m.					

GENERIC REFINERY - GULF COAST 100 kBPSP Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU		SHELL & TUBE EXCHANGER DATASHEET TEMA R / API 660 E-113 - BPA / MP steam generator (kettle)		DOC. No. CFU-000-ME-DS-004 REV / DATE A / 2026-10-02 STATUS Issued for review (FEED) PREPARED CLAUDE CODE CHK: - APP: - SHEET 12	
PERFORMANCE OF ONE UNIT					
1	Size / type (TEMA)	1300-6096 AKT (ID mm - tube length mm)	2	Connected in	1 parallel x 1 series
3	Surface per unit (total) / per shell	223 / 223 m2	4	Shells per unit	1
5	Service	BPA / MP steam generator (kettle)	6	Heat exchanged	4.91 MW
7	MTD (corrected)	56.4 C (approx.)	8	Transfer rate, service (clean: vendor)	450 W/m2K
SHELL SIDE / TUBE SIDE					
9	Fluid name (shell / tube)	BFW / MP steam / Bottom pumparound	10	Total flow (shell / tube)	8.9 / 165.7 t/h
11	Temperature in (shell / tube)	186 / 261 C	12	Temperature out (shell / tube)	186 / 224 C
13	Phase	Two-phase (condensing / boiling)	14	Physical properties	Per H&MB / vendor simulation
15	Allowable pressure drop (shell / tube)	0.35 / 0.7 bar	16	Fouling resistance (shell / tube)	0.0002 / 0.0004 m2K/W
CONSTRUCTION OF ONE SHELL					
17	Design pressure (shell / tube)	13 / 20 barg	18	Design temperature	290 C
19	Test pressure (shell / tube)	16.9 / 26.0 barg	20	Passes per shell (shell / tube)	1 / 2
21	Corrosion allowance	3.0 mm	22	Connections in/out (shell)	3" 300#
23	Connections in/out (tube)	8" 300#	24	Tube no. / OD / thk / length	627 / 19.05 mm / 14 BWG (2.11 mm) / 6096 mm
25	Tube pitch / layout	25.40 mm / 90 deg (square, cleanable)	26	Tube material	CS
27	Shell material / ID / thk	SA-516-70 / 1300 mm / 14 mm	28	Channel / bonnet thickness	12 mm
29	Tubesheet	Vendor (TEMA R / ASME UHX)	30	Floating head	-
31	Baffles	Single segmental, cut 25 % (V) - spacing vendor	32	Impingement protection	Plate (rho.v2 > 2230)
33	Expansion joint	None	34	Gaskets	Double-jacketed / spiral-wound
35	Code requirements	ASME VIII-1 / TEMA R / API 660	36	Bundle pull / removable	Yes
37	Weight per shell empty / water filled	10.3 / 20.2 t	38	Bundle weight (est.)	8.0 t
NOTES					
1. TEMA datasheet layout. Thermal design (HTRI) by vendor - geometry here is an estimate from area and the tube-count correlation. 2. Crude shell-side exchangers: square pitch, removable bundle, 1 m min. tube velocity (tube side) / no dead zones. 3. Stacking: max. 2 shells high; bundle-pull clearance = tube length + 1.5 m.					

GENERIC REFINERY - GULF COAST 100 kBPST Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: VDU		SHELL & TUBE EXCHANGER DATASHEET TEMA R / API 660 E-201 - VR / LP steam generator (kettle)		DOC. No. CFU-000-ME-DS-004 REV / DATE A / 2026-10-02 STATUS Issued for review (FEED) PREPARED CLAUDE CODE CHK: - APP: - SHEET 13	
PERFORMANCE OF ONE UNIT					
1	Size / type (TEMA)	900-6096 AKT (ID mm - tube length mm)	2	Connected in	1 parallel x 1 series
3	Surface per unit (total) / per shell	84 / 84 m2	4	Shells per unit	1
5	Service	VR / LP steam generator (kettle)	6	Heat exchanged	1.78 MW
7	MTD (corrected)	118.3 C (approx.)	8	Transfer rate, service (clean: vendor)	200 W/m2K
SHELL SIDE / TUBE SIDE					
9	Fluid name (shell / tube)	BFW / LP steam / Vacuum residue	10	Total flow (shell / tube)	3.0 / 108.3 t/h
11	Temperature in (shell / tube)	148 / 278 C	12	Temperature out (shell / tube)	148 / 255 C
13	Phase	Two-phase (condensing / boiling)	14	Physical properties	Per H&MB / vendor simulation
15	Allowable pressure drop (shell / tube)	0.35 / 0.7 bar	16	Fouling resistance (shell / tube)	0.0002 / 0.0009 m2K/W
CONSTRUCTION OF ONE SHELL					
17	Design pressure (shell / tube)	6 / 20 barg	18	Design temperature	310 C
19	Test pressure (shell / tube)	7.8 / 26.0 barg	20	Passes per shell (shell / tube)	1 / 2
21	Corrosion allowance	3.0 mm	22	Connections in/out (shell)	3" 150#
23	Connections in/out (tube)	8" 300#	24	Tube no. / OD / thk / length	236 / 19.05 mm / 14 BWG (2.11 mm) / 6096 mm
25	Tube pitch / layout	25.40 mm / 90 deg (square, cleanable)	26	Tube material	5Cr tubes
27	Shell material / ID / thk	SA-516-70 / 900 mm / 12 mm	28	Channel / bonnet thickness	10 mm
29	Tubesheet	Vendor (TEMA R / ASME UHX)	30	Floating head	-
31	Baffles	Single segmental, cut 25 % (V) - spacing vendor	32	Impingement protection	Plate (rho.v2 > 2230)
33	Expansion joint	None	34	Gaskets	Double-jacketed / spiral-wound
35	Code requirements	ASME VIII-1 / TEMA R / API 660	36	Bundle pull / removable	Yes
37	Weight per shell empty / water filled	4.9 / 9.6 t	38	Bundle weight (est.)	3.0 t
NOTES					
1. TEMA datasheet layout. Thermal design (HTRI) by vendor - geometry here is an estimate from area and the tube-count correlation. 2. Crude shell-side exchangers: square pitch, removable bundle, 1 m min. tube velocity (tube side) / no dead zones. 3. Stacking: max. 2 shells high; bundle-pull clearance = tube length + 1.5 m.					

GENERIC REFINERY - GULF COAST 100 kBPST Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU		SHELL & TUBE EXCHANGER DATASHEET TEMA R / API 660 E-114 - Stabiliser feed / bottoms		DOC. No. CFU-000-ME-DS-004	
				REV / DATE A / 2026-10-02	
				STATUS Issued for review (FEED)	
				PREPARED CLAUDE CODE CHK: - APP: -	
				SHEET 14	
PERFORMANCE OF ONE UNIT					
1	Size / type (TEMA)	650-6096 AES (ID mm - tube length mm)	2	Connected in	1 parallel x 1 series
3	Surface per unit (total) / per shell	86 / 86 m2	4	Shells per unit	1
5	Service	Stabiliser feed / bottoms	6	Heat exchanged	3.38 MW
7	MTD (corrected)	114.7 C (approx.)	8	Transfer rate, service (clean: vendor)	380 W/m2K
SHELL SIDE / TUBE SIDE					
9	Fluid name (shell / tube)	Unstab. naphtha / Stabiliser bottoms	10	Total flow (shell / tube)	114.9 / 108.4 t/h
11	Temperature in (shell / tube)	45 / 203 C	12	Temperature out (shell / tube)	91 / 163 C
13	Phase	L / L	14	Physical properties	Per H&MB / vendor simulation
15	Allowable pressure drop (shell / tube)	0.35 / 0.7 bar	16	Fouling resistance (shell / tube)	0.0004 / 0.0004 m2K/W
CONSTRUCTION OF ONE SHELL					
17	Design pressure (shell / tube)	20 / 20 barg	18	Design temperature	235 C
19	Test pressure (shell / tube)	26.0 / 26.0 barg	20	Passes per shell (shell / tube)	1 / 2
21	Corrosion allowance	3.0 mm	22	Connections in/out (shell)	8" 300#
23	Connections in/out (tube)	8" 300#	24	Tube no. / OD / thk / length	182 / 25.4 mm / 12 BWG (2.77 mm) / 6096 mm
25	Tube pitch / layout	31.75 mm / 90 deg (square, cleanable)	26	Tube material	CS
27	Shell material / ID / thk	SA-516-70 / 650 mm / 10 mm	28	Channel / bonnet thickness	10 mm
29	Tubesheet	Vendor (TEMA R / ASME UHX)	30	Floating head	Split-ring (S)
31	Baffles	Single segmental, cut 25 % (V) - spacing vendor	32	Impingement protection	Plate (rho.v2 > 2230)
33	Expansion joint	None	34	Gaskets	Double-jacketed / spiral-wound
35	Code requirements	ASME VIII-1 / TEMA R / API 660	36	Bundle pull / removable	Yes
37	Weight per shell empty / water filled	5.0 / 7.4 t	38	Bundle weight (est.)	2.3 t
NOTES					
1. TEMA datasheet layout. Thermal design (HTRI) by vendor - geometry here is an estimate from area and the tube-count correlation.					
2. Crude shell-side exchangers: square pitch, removable bundle, 1 m min. tube velocity (tube side) / no dead zones.					
3. Stacking: max. 2 shells high; bundle-pull clearance = tube length + 1.5 m.					

GENERIC REFINERY - GULF COAST 100 kBPST Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU		SHELL & TUBE EXCHANGER DATASHEET TEMA R / API 660 E-115 - Atm. overhead trim condenser		DOC. No. CFU-000-ME-DS-004	
				REV / DATE A / 2026-10-02	
				STATUS Issued for review (FEED)	
				PREPARED CLAUDE CODE CHK: - APP: -	
				SHEET 15	
PERFORMANCE OF ONE UNIT					
1	Size / type (TEMA)	900-6096 AEU (ID mm - tube length mm)	2	Connected in	1 parallel x 1 series
3	Surface per unit (total) / per shell	740 / 370 m2	4	Shells per unit	2
5	Service	Atm. overhead trim condenser	6	Heat exchanged	5.96 MW
7	MTD (corrected)	15.0 C (approx.)	8	Transfer rate, service (clean: vendor)	600 W/m2K
SHELL SIDE / TUBE SIDE					
9	Fluid name (shell / tube)	OH vapour/condensate / Cooling water	10	Total flow (shell / tube)	229.2 / 466.3 t/h
11	Temperature in (shell / tube)	60 / 32 C	12	Temperature out (shell / tube)	45 / 43 C
13	Phase	Two-phase (condensing / boiling)	14	Physical properties	Per H&MB / vendor simulation
15	Allowable pressure drop (shell / tube)	0.35 / 0.7 bar	16	Fouling resistance (shell / tube)	0.0004 / 0.00035 m2K/W
CONSTRUCTION OF ONE SHELL					
17	Design pressure (shell / tube)	3.5 / 7 barg	18	Design temperature	90 C
19	Test pressure (shell / tube)	4.5 / 9.1 barg	20	Passes per shell (shell / tube)	1 / 2
21	Corrosion allowance	3.0 mm	22	Connections in/out (shell)	10" 150#
23	Connections in/out (tube)	14" 150#	24	Tube no. / OD / thk / length	1040 / 19.05 mm / 14 BWG (2.11 mm) / 6096 mm
25	Tube pitch / layout	23.81 mm / 30 deg	26	Tube material	Ti tubes
27	Shell material / ID / thk	SA-516-70 / 900 mm / 12 mm	28	Channel / bonnet thickness	12 mm
29	Tubesheet	Vendor (TEMA R / ASME UHX)	30	Floating head	U-tube
31	Baffles	Single segmental, cut 25 % (V) - spacing vendor	32	Impingement protection	Plate (rho.v2 > 2230)
33	Expansion joint	None	34	Gaskets	Double-jacketed / spiral-wound
35	Code requirements	ASME VIII-1 / TEMA R / API 660	36	Bundle pull / removable	Yes
37	Weight per shell empty / water filled	9.0 / 13.5 t	38	Bundle weight (est.)	13.3 t
NOTES					
1. TEMA datasheet layout. Thermal design (HTRI) by vendor - geometry here is an estimate from area and the tube-count correlation.					
2. Crude shell-side exchangers: square pitch, removable bundle, 1 m min. tube velocity (tube side) / no dead zones.					
3. Stacking: max. 2 shells high; bundle-pull clearance = tube length + 1.5 m.					

GENERIC REFINERY - GULF COAST 100 kBPST Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU		SHELL & TUBE EXCHANGER DATASHEET TEMA R / API 660 E-116 - Stabiliser reboiler (HP steam)		DOC. No. CFU-000-ME-DS-004 REV / DATE A / 2026-10-02 STATUS Issued for review (FEED) PREPARED CLAUDE CODE CHK: - APP: - SHEET 16	
PERFORMANCE OF ONE UNIT					
1	Size / type (TEMA)	1300-6096 BKT (ID mm - tube length mm)	2	Connected in	1 parallel x 1 series
3	Surface per unit (total) / per shell	242 / 242 m2	4	Shells per unit	1
5	Service	Stabiliser reboiler (HP steam)	6	Heat exchanged	10.83 MW
7	MTD (corrected)	55.4 C (approx.)	8	Transfer rate, service (clean: vendor)	900 W/m2K
SHELL SIDE / TUBE SIDE					
9	Fluid name (shell / tube)	Naphtha / HP steam	10	Total flow (shell / tube)	129.9 / 22.9 t/h
11	Temperature in (shell / tube)	193 / 253 C	12	Temperature out (shell / tube)	203 / 253 C
13	Phase	Two-phase (condensing / boiling)	14	Physical properties	Per H&MB / vendor simulation
15	Allowable pressure drop (shell / tube)	0.35 / 0.7 bar	16	Fouling resistance (shell / tube)	0.0004 / 0.0001 m2K/W
CONSTRUCTION OF ONE SHELL					
17	Design pressure (shell / tube)	15 / 45 barg	18	Design temperature	430 C
19	Test pressure (shell / tube)	19.5 / 58.5 barg	20	Passes per shell (shell / tube)	1 / 2
21	Corrosion allowance	3.0 mm	22	Connections in/out (shell)	8" 300#
23	Connections in/out (tube)	3" 600#	24	Tube no. / OD / thk / length	680 / 19.05 mm / 14 BWG (2.11 mm) / 6096 mm
25	Tube pitch / layout	25.40 mm / 90 deg (square, cleanable)	26	Tube material	CS
27	Shell material / ID / thk	SA-516-70 / 1300 mm / 18 mm	28	Channel / bonnet thickness	32 mm
29	Tubesheet	Vendor (TEMA R / ASME UHX)	30	Floating head	-
31	Baffles	Single segmental, cut 25 % (V) - spacing vendor	32	Impingement protection	Plate (rho.v2 > 2230)
33	Expansion joint	None	34	Gaskets	Double-jacketed / spiral-wound
35	Code requirements	ASME VIII-1 / TEMA R / API 660	36	Bundle pull / removable	Yes
37	Weight per shell empty / water filled	12.9 / 22.8 t	38	Bundle weight (est.)	8.7 t
NOTES					
1. TEMA datasheet layout. Thermal design (HTRI) by vendor - geometry here is an estimate from area and the tube-count correlation. 2. Crude shell-side exchangers: square pitch, removable bundle, 1 m min. tube velocity (tube side) / no dead zones. 3. Stacking: max. 2 shells high; bundle-pull clearance = tube length + 1.5 m.					

GENERIC REFINERY - GULF COAST 100 kBPST Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU		SHELL & TUBE EXCHANGER DATASHEET TEMA R / API 660 E-117 - Splitter reboiler (MP steam)		DOC. No. CFU-000-ME-DS-004 REV / DATE A / 2026-10-02 STATUS Issued for review (FEED) PREPARED CLAUDE CODE CHK: - APP: - SHEET 17	
PERFORMANCE OF ONE UNIT					
1	Size / type (TEMA)	750-6096 BXM (ID mm - tube length mm)	2	Connected in	1 parallel x 1 series
3	Surface per unit (total) / per shell	225 / 225 m2	4	Shells per unit	1
5	Service	Splitter reboiler (MP steam)	6	Heat exchanged	6.83 MW
7	MTD (corrected)	37.7 C (approx.)	8	Transfer rate, service (clean: vendor)	900 W/m2K
SHELL SIDE / TUBE SIDE					
9	Fluid name (shell / tube)	Naphtha / MP steam	10	Total flow (shell / tube)	307.5 / 12.4 t/h
11	Temperature in (shell / tube)	144 / 186 C	12	Temperature out (shell / tube)	152 / 186 C
13	Phase	Two-phase (condensing / boiling)	14	Physical properties	Per H&MB / vendor simulation
15	Allowable pressure drop (shell / tube)	0.35 / 0.7 bar	16	Fouling resistance (shell / tube)	0.0004 / 0.0001 m2K/W
CONSTRUCTION OF ONE SHELL					
17	Design pressure (shell / tube)	5 / 13 barg	18	Design temperature	280 C
19	Test pressure (shell / tube)	6.5 / 16.9 barg	20	Passes per shell (shell / tube)	1 / 2
21	Corrosion allowance	3.0 mm	22	Connections in/out (shell)	12" 150#
23	Connections in/out (tube)	3" 300#	24	Tube no. / OD / thk / length	632 / 19.05 mm / 14 BWG (2.11 mm) / 6096 mm
25	Tube pitch / layout	25.40 mm / 90 deg (square, cleanable)	26	Tube material	CS
27	Shell material / ID / thk	SA-516-70 / 750 mm / 10 mm	28	Channel / bonnet thickness	10 mm
29	Tubesheet	Vendor (TEMA R / ASME UHX)	30	Floating head	-
31	Baffles	Single segmental, cut 25 % (V) - spacing vendor	32	Impingement protection	Plate (rho.v2 > 2230)
33	Expansion joint	None	34	Gaskets	Double-jacketed / spiral-wound
35	Code requirements	ASME VIII-1 / TEMA R / API 660	36	Bundle pull / removable	Fixed TS
37	Weight per shell empty / water filled	8.0 / 11.2 t	38	Bundle weight (est.)	8.1 t
NOTES					
1. TEMA datasheet layout. Thermal design (HTRI) by vendor - geometry here is an estimate from area and the tube-count correlation. 2. Crude shell-side exchangers: square pitch, removable bundle, 1 m min. tube velocity (tube side) / no dead zones. 3. Stacking: max. 2 shells high; bundle-pull clearance = tube length + 1.5 m.					

GENERIC REFINERY - GULF COAST 100 kBPST Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU		SHELL & TUBE EXCHANGER DATASHEET TEMA R / API 660 E-118 - Desalter wash water / brine		DOC. No. CFU-000-ME-DS-004	
				REV / DATE A / 2026-10-02	
				STATUS Issued for review (FEED)	
				PREPARED CLAUDE CODE CHK: - APP: -	
				SHEET 18	
PERFORMANCE OF ONE UNIT					
1	Size / type (TEMA)	600-6096 AEL (ID mm - tube length mm)	2	Connected in	1 parallel x 1 series
3	Surface per unit (total) / per shell	84 / 84 m2	4	Shells per unit	1
5	Service	Desalter wash water / brine	6	Heat exchanged	1.62 MW
7	MTD (corrected)	33.2 C (approx.)	8	Transfer rate, service (clean: vendor)	800 W/m2K
SHELL SIDE / TUBE SIDE					
9	Fluid name (shell / tube)	Wash water / Brine	10	Total flow (shell / tube)	33.1 / 33.8 t/h
11	Temperature in (shell / tube)	50 / 131 C	12	Temperature out (shell / tube)	121 / 106 C
13	Phase	L / L	14	Physical properties	Per H&MB / vendor simulation
15	Allowable pressure drop (shell / tube)	0.35 / 0.7 bar	16	Fouling resistance (shell / tube)	0.0004 / 0.0004 m2K/W
CONSTRUCTION OF ONE SHELL					
17	Design pressure (shell / tube)	20 / 20 barg	18	Design temperature	160 C
19	Test pressure (shell / tube)	26.0 / 26.0 barg	20	Passes per shell (shell / tube)	1 / 2
21	Corrosion allowance	3.0 mm	22	Connections in/out (shell)	4" 300#
23	Connections in/out (tube)	4" 300#	24	Tube no. / OD / thk / length	177 / 25.4 mm / 12 BWG (2.77 mm) / 6096 mm
25	Tube pitch / layout	31.75 mm / 90 deg (square, cleanable)	26	Tube material	duplex 2205 tubes
27	Shell material / ID / thk	SA-516-70 / 600 mm / 10 mm	28	Channel / bonnet thickness	10 mm
29	Tubesheet	Vendor (TEMA R / ASME UHX)	30	Floating head	-
31	Baffles	Single segmental, cut 25 % (V) - spacing vendor	32	Impingement protection	Plate (rho.v2 > 2230)
33	Expansion joint	None	34	Gaskets	Double-jacketed / spiral-wound
35	Code requirements	ASME VIII-1 / TEMA R / API 660	36	Bundle pull / removable	Fixed TS
37	Weight per shell empty / water filled	4.7 / 6.8 t	38	Bundle weight (est.)	2.3 t
NOTES					
1. TEMA datasheet layout. Thermal design (HTRI) by vendor - geometry here is an estimate from area and the tube-count correlation.					
2. Crude shell-side exchangers: square pitch, removable bundle, 1 m min. tube velocity (tube side) / no dead zones.					
3. Stacking: max. 2 shells high; bundle-pull clearance = tube length + 1.5 m.					

GENERIC REFINERY - GULF COAST 100 kBPST Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: VDU		SHELL & TUBE EXCHANGER DATASHEET TEMA R / API 660 E-202 - 1st-stage ejector intercondenser		DOC. No. CFU-000-ME-DS-004 REV / DATE A / 2026-10-02 STATUS Issued for review (FEED) PREPARED CLAUDE CODE CHK: - APP: - SHEET 19	
PERFORMANCE OF ONE UNIT					
1	Size / type (TEMA)	1100-6096 AXS (ID mm - tube length mm)	2	Connected in	1 parallel x 1 series
3	Surface per unit (total) / per shell	549 / 549 m2	4	Shells per unit	1
5	Service	1st-stage ejector intercondenser	6	Heat exchanged	6.95 MW
7	MTD (corrected)	17.5 C (approx.)	8	Transfer rate, service (clean: vendor)	900 W/m2K
SHELL SIDE / TUBE SIDE					
9	Fluid name (shell / tube)	Steam / NCG / Cooling water	10	Total flow (shell / tube)	10.9 / 543.8 t/h
11	Temperature in (shell / tube)	70 / 32 C	12	Temperature out (shell / tube)	40 / 43 C
13	Phase	Two-phase (condensing / boiling)	14	Physical properties	Per H&MB / vendor simulation
15	Allowable pressure drop (shell / tube)	0.35 / 0.7 bar	16	Fouling resistance (shell / tube)	0.0004 / 0.00035 m2K/W
CONSTRUCTION OF ONE SHELL					
17	Design pressure (shell / tube)	FV-3.5 / 7 barg	18	Design temperature	100 C
19	Test pressure (shell / tube)	4.5 / 9.1 barg	20	Passes per shell (shell / tube)	1 / 1
21	Corrosion allowance	3.0 mm	22	Connections in/out (shell)	3" 150#
23	Connections in/out (tube)	16" 150#	24	Tube no. / OD / thk / length	1543 / 19.05 mm / 14 BWG (2.11 mm) / 6096 mm
25	Tube pitch / layout	23.81 mm / 30 deg	26	Tube material	304L tubes
27	Shell material / ID / thk	SA-516-70 / 1100 mm / 14 mm	28	Channel / bonnet thickness	14 mm
29	Tubesheet	Vendor (TEMA R / ASME UHX)	30	Floating head	-
31	Baffles	Single segmental, cut 25 % (V) - spacing vendor	32	Impingement protection	Plate (rho.v2 > 2230)
33	Expansion joint	None	34	Gaskets	Double-jacketed / spiral-wound
35	Code requirements	ASME VIII-1 / TEMA R / API 660	36	Bundle pull / removable	Yes
37	Weight per shell empty / water filled	18.6 / 25.4 t	38	Bundle weight (est.)	19.7 t
NOTES					
1. TEMA datasheet layout. Thermal design (HTRI) by vendor - geometry here is an estimate from area and the tube-count correlation. 2. Crude shell-side exchangers: square pitch, removable bundle, 1 m min. tube velocity (tube side) / no dead zones. 3. Stacking: max. 2 shells high; bundle-pull clearance = tube length + 1.5 m.					

GENERIC REFINERY - GULF COAST 100 kBPST Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: VDU		SHELL & TUBE EXCHANGER DATASHEET TEMA R / API 660 E-203 - 2nd-stage ejector intercondenser		DOC. No. CFU-000-ME-DS-004 REV / DATE A / 2026-10-02 STATUS Issued for review (FEED) PREPARED CLAUDE CODE CHK: - APP: - SHEET 20	
PERFORMANCE OF ONE UNIT					
1	Size / type (TEMA)	500-6096 AXS (ID mm - tube length mm)	2	Connected in	1 parallel x 1 series
3	Surface per unit (total) / per shell	79 / 79 m2	4	Shells per unit	1
5	Service	2nd-stage ejector intercondenser	6	Heat exchanged	1.00 MW
7	MTD (corrected)	17.5 C (approx.)	8	Transfer rate, service (clean: vendor)	900 W/m2K
SHELL SIDE / TUBE SIDE					
9	Fluid name (shell / tube)	Steam / NCG / Cooling water	10	Total flow (shell / tube)	1.6 / 78.0 t/h
11	Temperature in (shell / tube)	70 / 32 C	12	Temperature out (shell / tube)	40 / 43 C
13	Phase	Two-phase (condensing / boiling)	14	Physical properties	Per H&MB / vendor simulation
15	Allowable pressure drop (shell / tube)	0.35 / 0.7 bar	16	Fouling resistance (shell / tube)	0.0004 / 0.00035 m2K/W
CONSTRUCTION OF ONE SHELL					
17	Design pressure (shell / tube)	FV-3.5 / 7 barg	18	Design temperature	100 C
19	Test pressure (shell / tube)	4.5 / 9.1 barg	20	Passes per shell (shell / tube)	1 / 1
21	Corrosion allowance	3.0 mm	22	Connections in/out (shell)	3" 150#
23	Connections in/out (tube)	6" 150#	24	Tube no. / OD / thk / length	222 / 19.05 mm / 14 BWG (2.11 mm) / 6096 mm
25	Tube pitch / layout	23.81 mm / 30 deg	26	Tube material	304L tubes
27	Shell material / ID / thk	SA-516-70 / 500 mm / 10 mm	28	Channel / bonnet thickness	10 mm
29	Tubesheet	Vendor (TEMA R / ASME UHX)	30	Floating head	-
31	Baffles	Single segmental, cut 25 % (V) - spacing vendor	32	Impingement protection	Plate (rho.v2 > 2230)
33	Expansion joint	None	34	Gaskets	Double-jacketed / spiral-wound
35	Code requirements	ASME VIII-1 / TEMA R / API 660	36	Bundle pull / removable	Yes
37	Weight per shell empty / water filled	3.5 / 5.0 t	38	Bundle weight (est.)	2.8 t
NOTES					
1. TEMA datasheet layout. Thermal design (HTRI) by vendor - geometry here is an estimate from area and the tube-count correlation. 2. Crude shell-side exchangers: square pitch, removable bundle, 1 m min. tube velocity (tube side) / no dead zones. 3. Stacking: max. 2 shells high; bundle-pull clearance = tube length + 1.5 m.					

GENERIC REFINERY - GULF COAST 100 kBPST Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: VDU		SHELL & TUBE EXCHANGER DATASHEET TEMA R / API 660 E-204 - Ejector aftercondenser		DOC. No. CFU-000-ME-DS-004 REV / DATE A / 2026-10-02 STATUS Issued for review (FEED) PREPARED CLAUDE CODE CHK: - APP: - SHEET 21	
PERFORMANCE OF ONE UNIT					
1	Size / type (TEMA)	400-4877 AXS (ID mm - tube length mm)	2	Connected in	1 parallel x 1 series
3	Surface per unit (total) / per shell	30 / 30 m2	4	Shells per unit	1
5	Service	Ejector aftercondenser	6	Heat exchanged	0.38 MW
7	MTD (corrected)	17.5 C (approx.)	8	Transfer rate, service (clean: vendor)	900 W/m2K
SHELL SIDE / TUBE SIDE					
9	Fluid name (shell / tube)	Steam / NCG / Cooling water	10	Total flow (shell / tube)	0.6 / 30.0 t/h
11	Temperature in (shell / tube)	70 / 32 C	12	Temperature out (shell / tube)	40 / 43 C
13	Phase	Two-phase (condensing / boiling)	14	Physical properties	Per H&MB / vendor simulation
15	Allowable pressure drop (shell / tube)	0.35 / 0.7 bar	16	Fouling resistance (shell / tube)	0.0004 / 0.00035 m2K/W
CONSTRUCTION OF ONE SHELL					
17	Design pressure (shell / tube)	FV-3.5 / 7 barg	18	Design temperature	100 C
19	Test pressure (shell / tube)	4.5 / 9.1 barg	20	Passes per shell (shell / tube)	1 / 1
21	Corrosion allowance	3.0 mm	22	Connections in/out (shell)	3" 150#
23	Connections in/out (tube)	4" 150#	24	Tube no. / OD / thk / length	107 / 19.05 mm / 14 BWG (2.11 mm) / 4877 mm
25	Tube pitch / layout	23.81 mm / 30 deg	26	Tube material	304L tubes
27	Shell material / ID / thk	SA-516-70 / 400 mm / 10 mm	28	Channel / bonnet thickness	10 mm
29	Tubesheet	Vendor (TEMA R / ASME UHX)	30	Floating head	-
31	Baffles	Single segmental, cut 25 % (V) - spacing vendor	32	Impingement protection	Plate (rho.v2 > 2230)
33	Expansion joint	None	34	Gaskets	Double-jacketed / spiral-wound
35	Code requirements	ASME VIII-1 / TEMA R / API 660	36	Bundle pull / removable	Yes
37	Weight per shell empty / water filled	1.9 / 2.7 t	38	Bundle weight (est.)	1.1 t
NOTES					
1. TEMA datasheet layout. Thermal design (HTRI) by vendor - geometry here is an estimate from area and the tube-count correlation. 2. Crude shell-side exchangers: square pitch, removable bundle, 1 m min. tube velocity (tube side) / no dead zones. 3. Stacking: max. 2 shells high; bundle-pull clearance = tube length + 1.5 m.					